Discussion 08

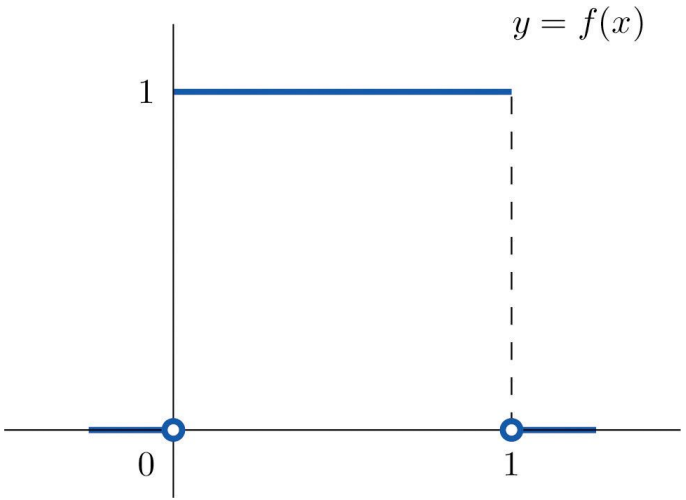
Monte Carlo & Global Illumination

Computer Graphics and Imaging
UC Berkeley CS 184

Inversion Method

Inversion Method

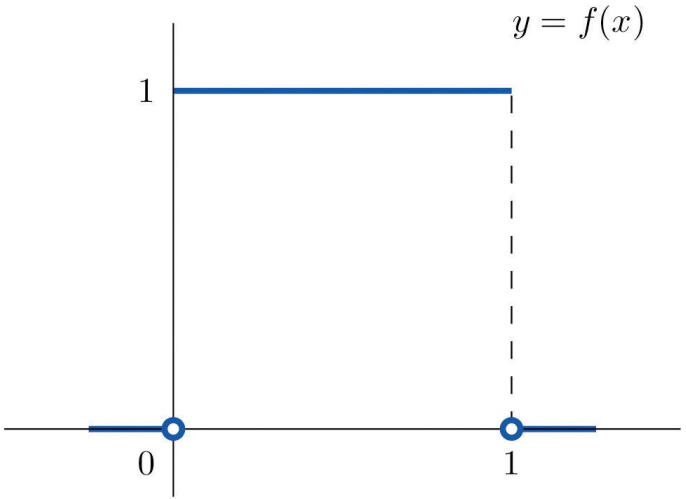
- Say our computer can sample values for *U*, which is *Uniform[0, 1]*.
- How to sample values from any PDF, *p(x)*?



Inversion Method

- Say our computer can sample values for *U*, which is *Uniform[0, 1]*.
- How to sample values from any PDF, *p(x)*?

1. Calculate the CDF, *F(x)*, by integrating *p(x)*.
2. Invert the CDF → $F^{-1}(x)$.
3. Sample U → X = F^{-1}(U) to sample from p(x)

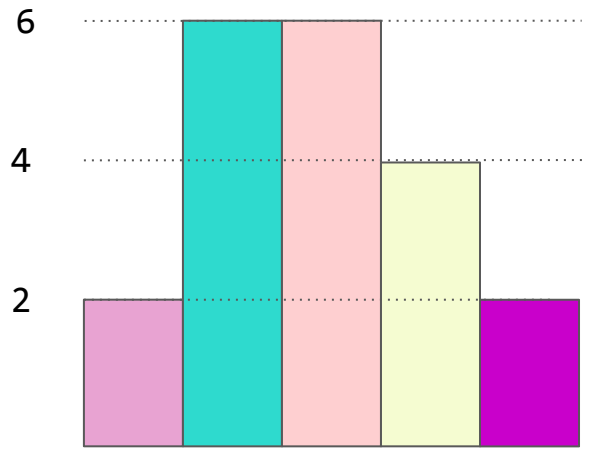


Inversion Method

Benefit of the inversion method: we can generate a probability distribution using uniform random numbers!

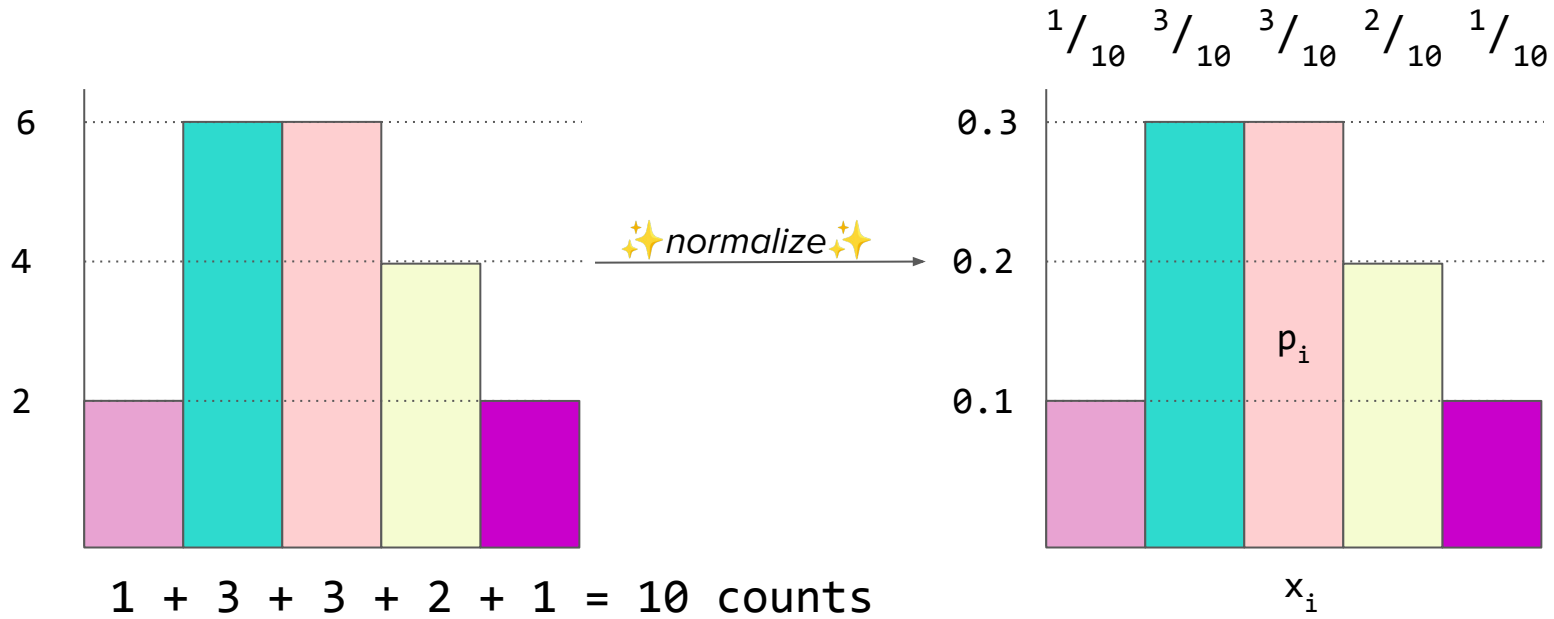
Steps

1. Calculate the CDF, $F(x)$, by integrating $p(x)$.
2. Invert the CDF ➔ $F^{-1}(x)$.
3. Sample U ➔ $X = F^{-1}(U)$ to sample *from* $p(x)$



Inversion Method

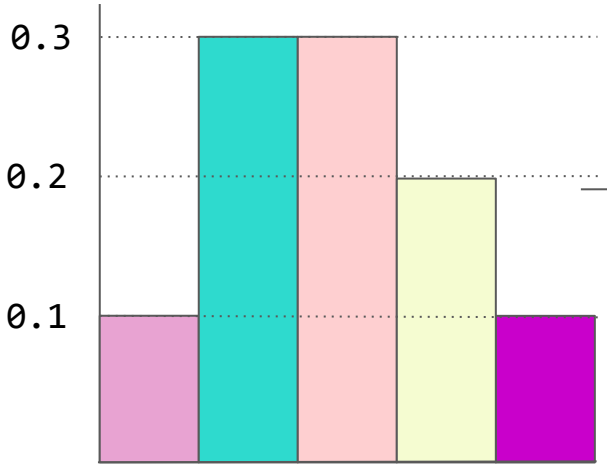
Compute PDF (probability density function) by normalizing



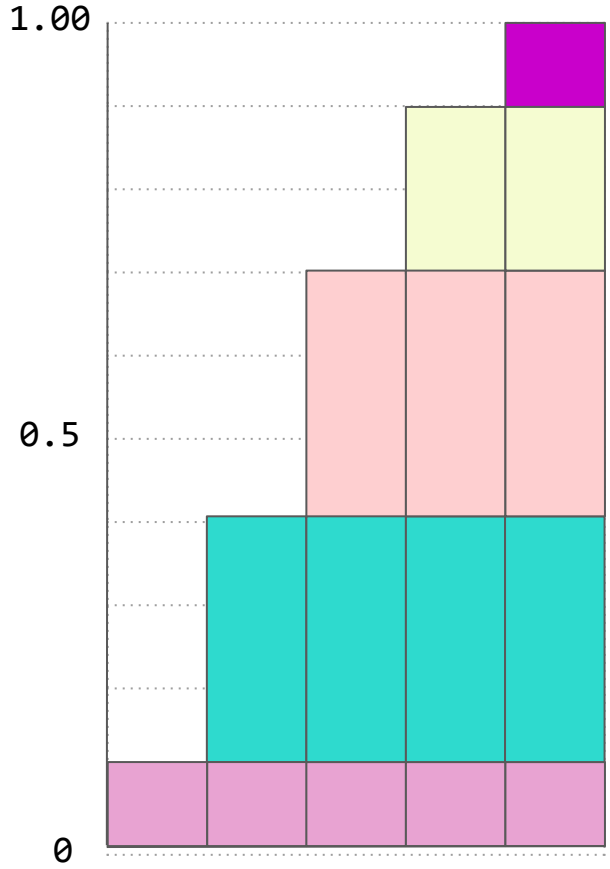
Inversion Method

Compute CDF (cumulative density function)

`P(x) = Pr(X < x)`



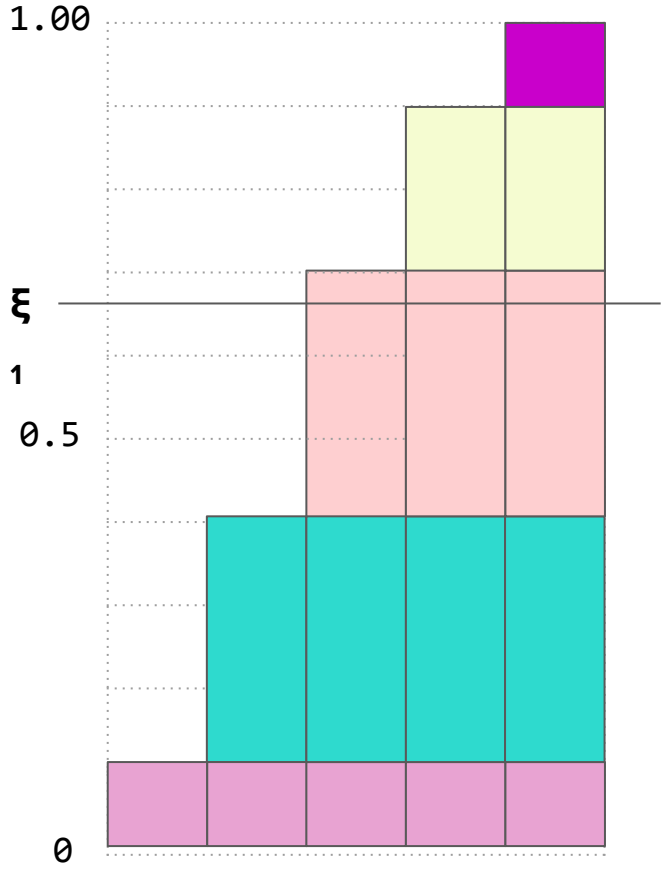
compute CDF from PDF



Inversion Method

Step 3 - Invert the CDF, and solve for $F^{-1}(\xi)$

$F^{-1}(\xi_1)$ =

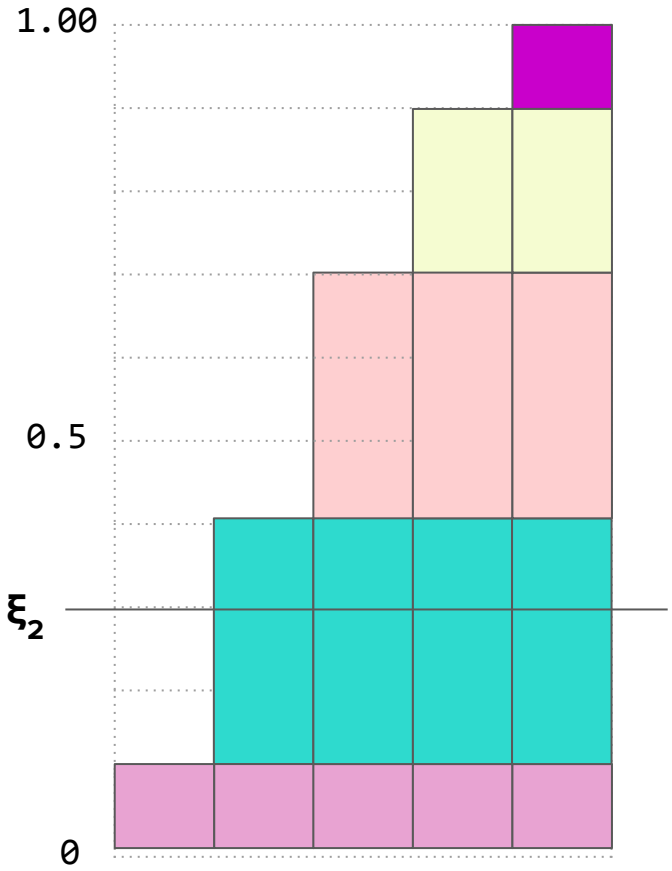


Inversion Method

Step 3 - Invert the CDF, and solve for $F^{-1}(\xi)$

$F^{-1}(\xi_1)$ =

$F^{-1}(\xi_2)$ =



Worksheet Question 1

1. Inversion Method

Given a uniform random variable U in the interval $[0, 1]$, we can generate a random variable from any other one dimensional distribution using its cumulative distribution function: $X = F^{-1}(U)$. This is how we choose sample points when running a ray tracing algorithm.

1. What function of U will return a sample from the exponential distribution (with parameter λ)? This distribution has density $p_\lambda(x) = \lambda e^{-\lambda x}$, and is defined for $x \geq 0$.

As a reminder, the steps of the inversion method are

1. Compute CDF (cumulative density function), by integrating the PDF
2. Invert the CDF ➜ $F^{-1}(x)$.
3. Sampling X according to $p(x)$ is achieved by sampling U ➜ $X = F^{-1}(U)$.

1. Inversion Method

Given a uniform random variable U in the interval $[0, 1]$, we can generate a random variable from any other one dimensional distribution using its cumulative distribution function: $X = F^{-1}(U)$. This is how we choose sample points when running a ray tracing algorithm.

1. What function of U will return a sample from the exponential distribution (with parameter λ)? This distribution has density $p_\lambda(x) = \lambda e^{-\lambda x}$, and is defined for $x \geq 0$.

Solution: First, we need to calculate the CDF:

$$F_\lambda(x) = \int_0^x p_\lambda(t)dt = \int_0^x \lambda e^{-\lambda t}dt = -e^{-\lambda t}\Big|_0^x = 1 - e^{-\lambda x}$$

Set the CDF equal to U:

$$U = 1 - e^{-\lambda x}$$

Solving for x:

$$e^{-\lambda x} = 1 - U$$
$$-\lambda x = \ln(1 - U)$$
$$x = -\frac{\ln(1 - U)}{\lambda}$$

Thus, the inverse function is given by:

$$F_\lambda^{-1}(x) = -\frac{\ln(1 - x)}{\lambda}$$

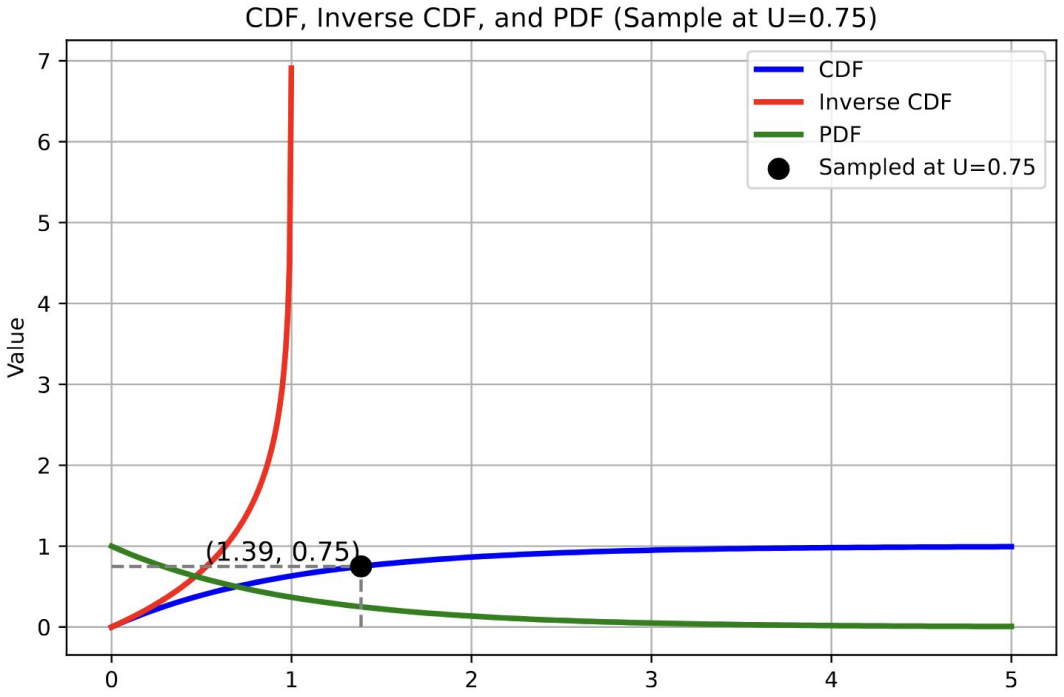
and we can return:

$$-\frac{\ln(1 - U)}{\lambda}$$

to sample from the exponential distribution.

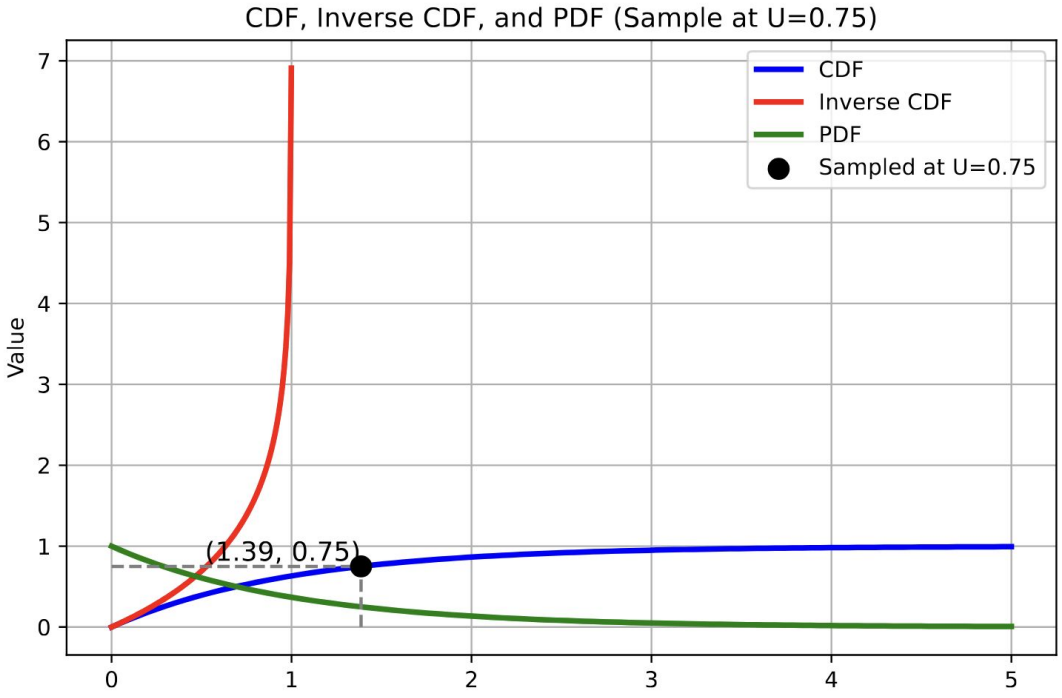
1. Inversion Method

2. For $\lambda = 1$, the plot below shows the PDF and CDF. If we sample $U = 0.75$, determine the corresponding sampled value using the inverse method. Mark the corresponding point on the CDF curve in the graph.



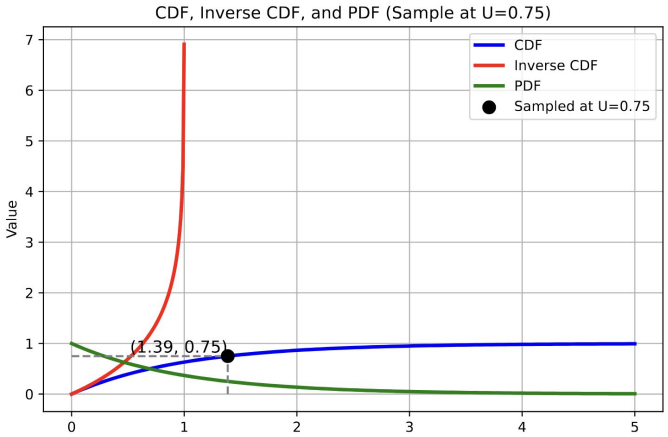
1. Inversion Method

3. What does the x-axis represent for the blue CDF curve?



1. Inversion Method

3. What does the x-axis represent for the blue CDF curve?



Solution: The x-axis represents the possible values that can be sampled from the exponential distribution. Given a uniform random variable U, the CDF maps it to a corresponding sampled value x from the distribution. The x-axis, therefore, represents the range of values we could obtain when transforming a uniform sample using the inverse CDF.

Monte Carlo Integration

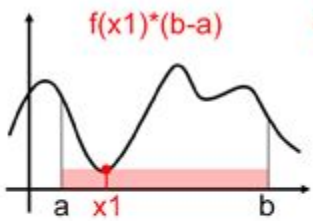
Monte Carlo Integration

- Estimate the value of an integral: $\int_a^b f(x)dx$
- Take *n* samples of *f(x)*, where *x* is randomly chosen from between *a* and *b*.



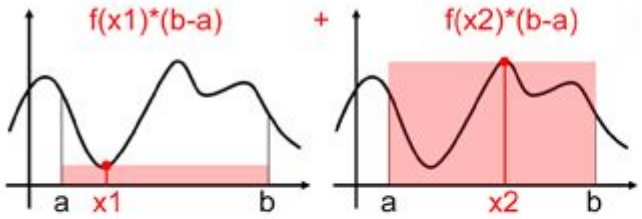
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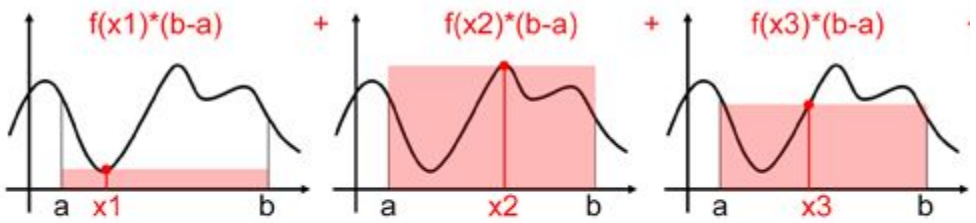
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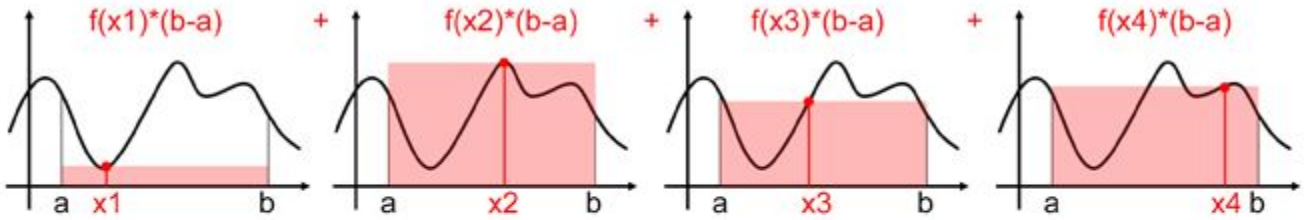
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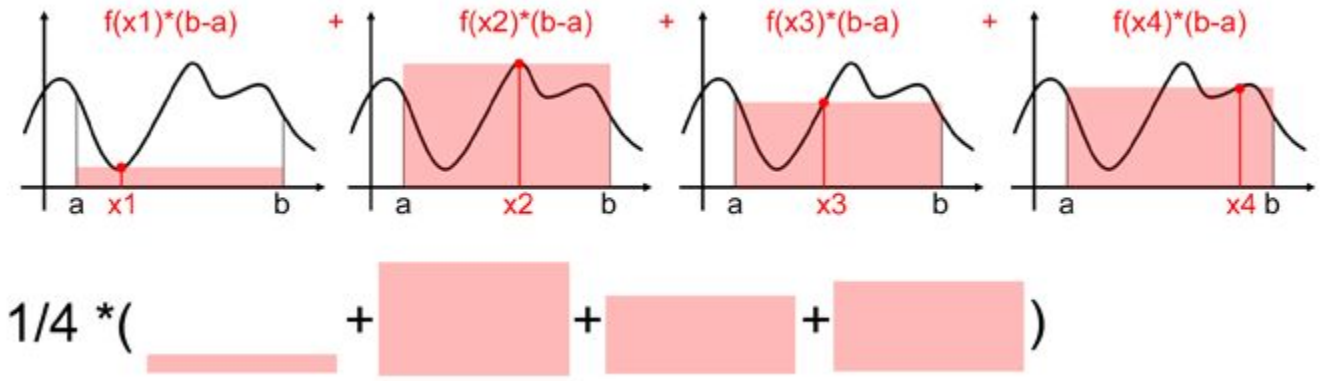
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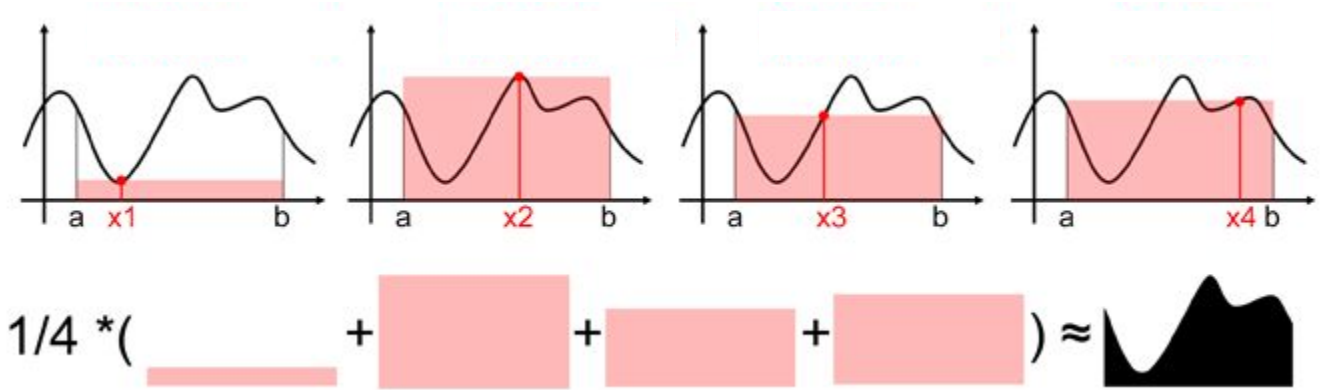
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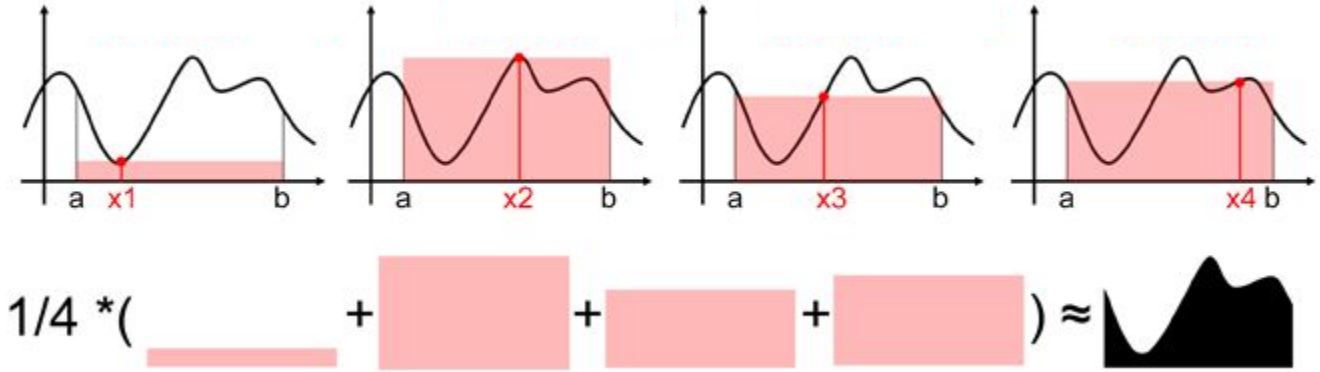
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Monte Carlo Integration

- Estimate the value of an integral: $\int_a^b f(x)dx$
- Take *n* samples of *f(x)*, where *x* is randomly chosen from between *a* and *b*.
- This estimate is <u>unbiased</u> → $E[F_n] = \int_a^b f(x)dx$



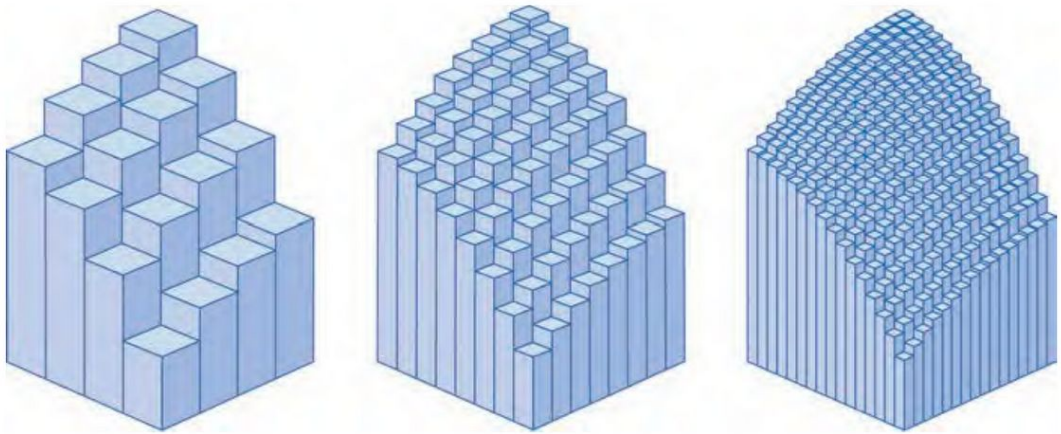
Why not Riemann sums?

- Hard to approximate complex shapes using rectangles.



Why not Riemann sums?

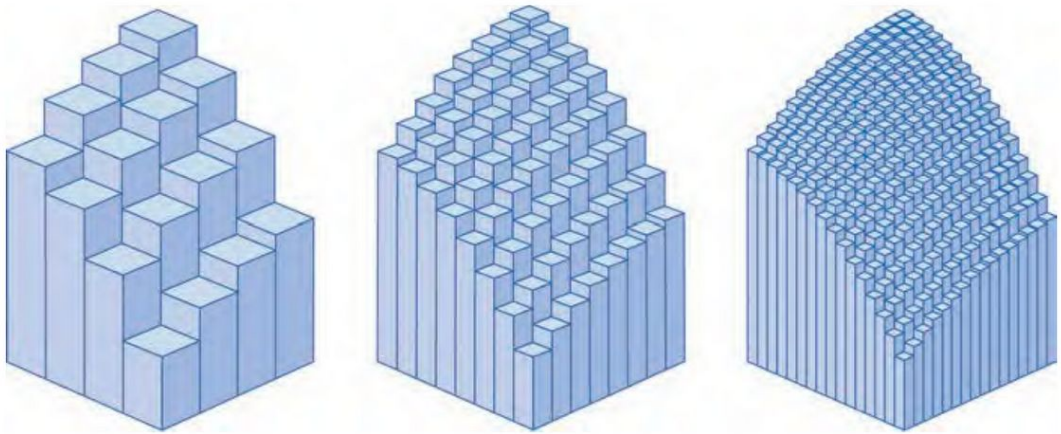
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- Need lots of samples to get a good estimate! (Curse of dimensionality.)



Why not Riemann sums?

- Hard to approximate complex shapes using rectangles.
- Need lots of samples to get a good estimate! (Curse of dimensionality.)
- Monte Carlo sampling is easy to implement.

```
random_uniform()
```



Monte Carlo Integration

Let us define the Monte Carlo estimator for the definite integral of given function $f(x)$

Definite integral

$$\int_a^b f(x)dx$$

Random variable

$$X_i \sim p(x)$$

p(x) can be any distribution! (Not just uniform.)

Monte Carlo estimator

$$F_N = \frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{p(X_i)}$$

Question 2

2. Unbiased Estimators

1. Let $f : [-2, 2] \times [-2, 2] \to \mathbb{R}$ be a function. You have a machine that allows you to sample $2n$ values independently and uniformly from the interval $[-2, 2]$. Construct an unbiased Monte Carlo estimator for

$$F = \int_{-2}^{2} \int_{-2}^{2} f(x, y) dx dy.$$

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REMINDER:

Monte Carlo estimator $$F_N = \frac{1}{N} \sum_{i=1}^{N} \frac{f(X_i)}{p(X_i)}$$

Where N = number of samples, X_i: ith random variable sampled from distribution, p(x) is the PDF of distribution for sampling values for the domain, and f(x) is our function (our range) we would like to estimate

2. Unbiased Estimators

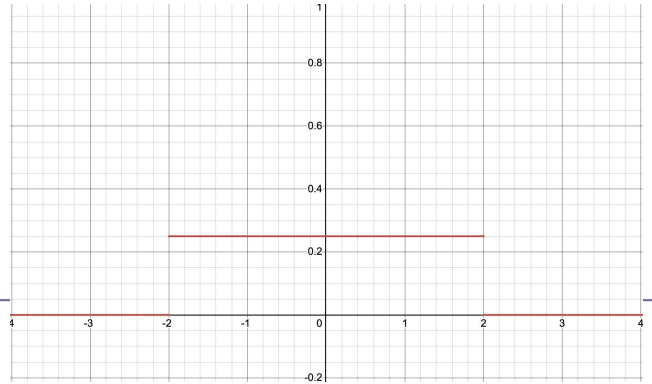
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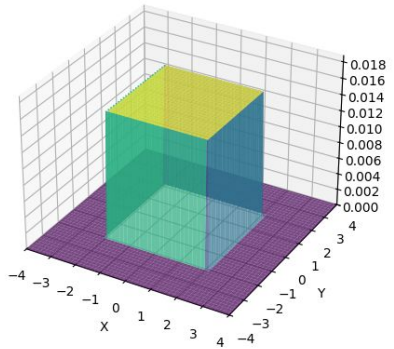
Solution: Draw n random samples $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$. Then we take

$$\langle F_n \rangle := \frac{1}{n} \sum_{i=1}^{n} \frac{f(X_i, Y_i)}{p(X_i, Y_i)} \text{ , where } p(X_i, Y_i) = p(X_i)p(Y_i) = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16}$$

p(x):



p(x, y):



2. Unbiased Estimators

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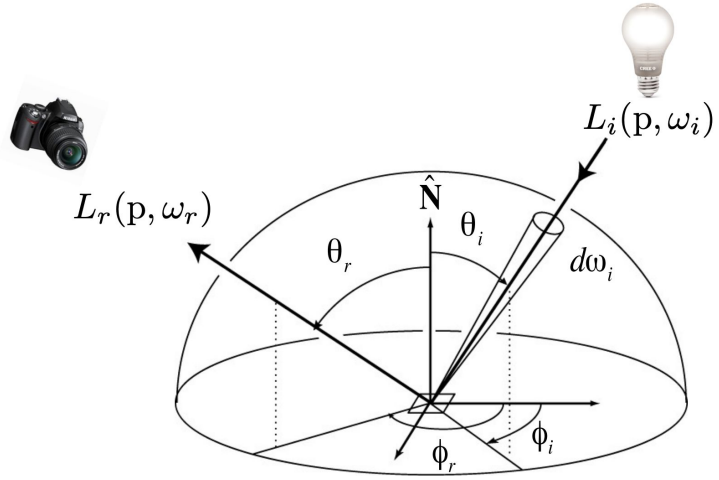
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&= \frac{1}{n} \sum_{i=1}^{n} \frac{f(X_i, Y_i)}{\frac{1}{16}} \\
&= \frac{16}{n} \sum_{i=1}^{n} f(X_i, Y_i).
\end{aligned}$$

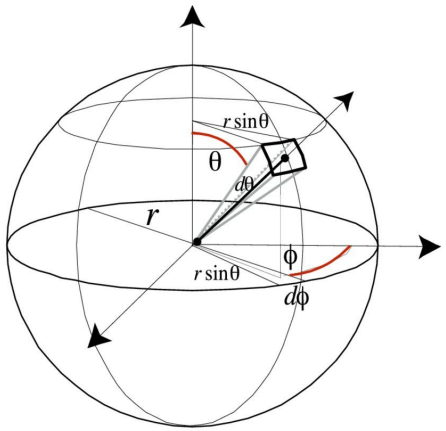
Light Transport

Light Transport

- ω_i is the "direction" of <u>incoming</u> light.
- ω_o is the "direction" of <u>outgoing</u> light.



- The BRDF, $f_r(p, \omega_i, \omega_o)$, gives the amount of light from ω_i that is reflected in the direction ω_o.



$$\begin{aligned} \mathrm{d}A &= (r\,\mathrm{d}\theta)(r\sin\theta\,\mathrm{d}\phi) \\ &= r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi \end{aligned}$$

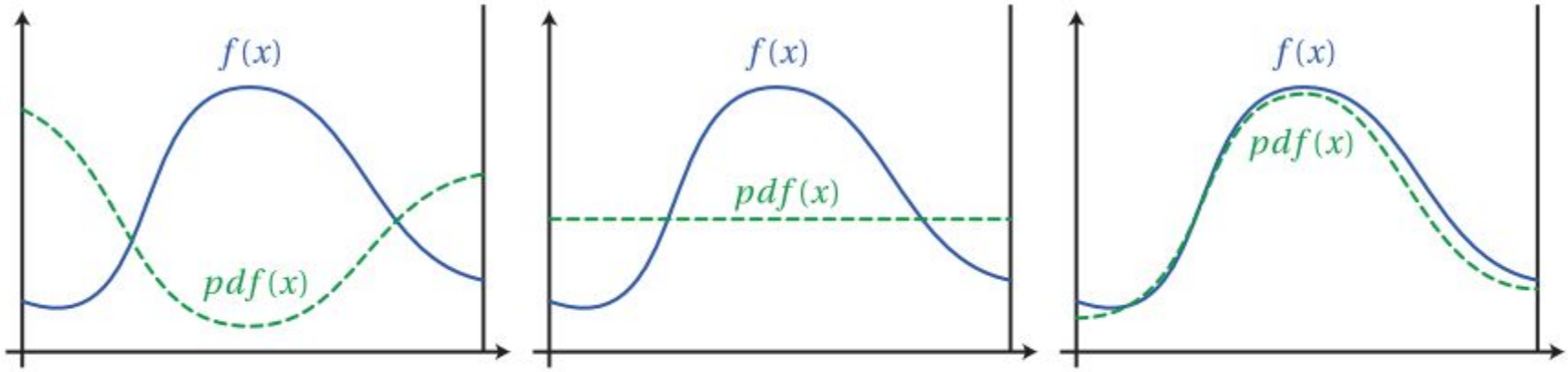
$$\mathrm{d}\omega = \frac{\mathrm{d}A}{r^2} = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$$

Question 3

6. Conceptually, why does cosine-weighted hemisphere sampling outperform uniform sampling over a hemisphere?

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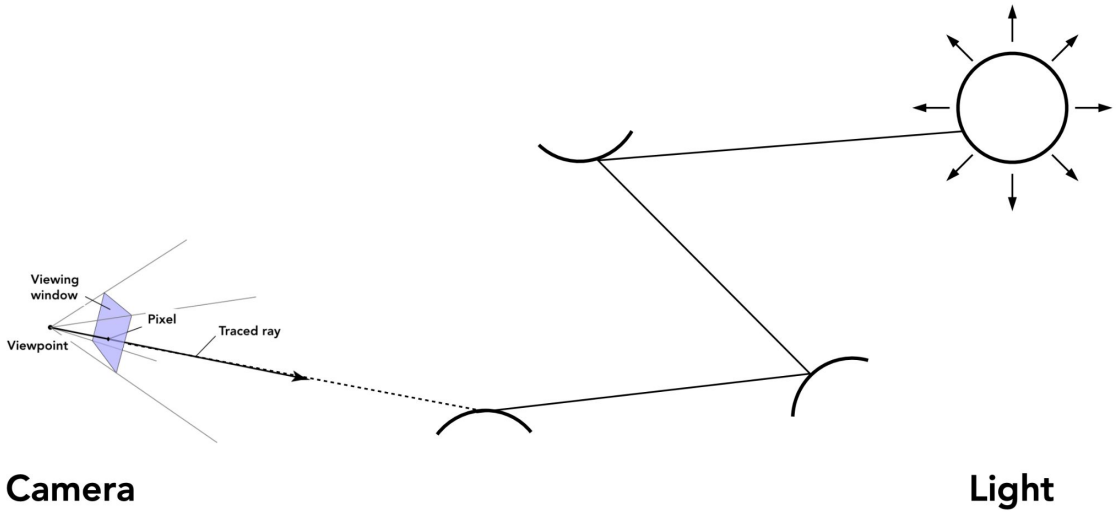
- **Importance sampling → want PDF of *X*'s to resemble integrand.**
- **Why does this work?**



Global Illumination

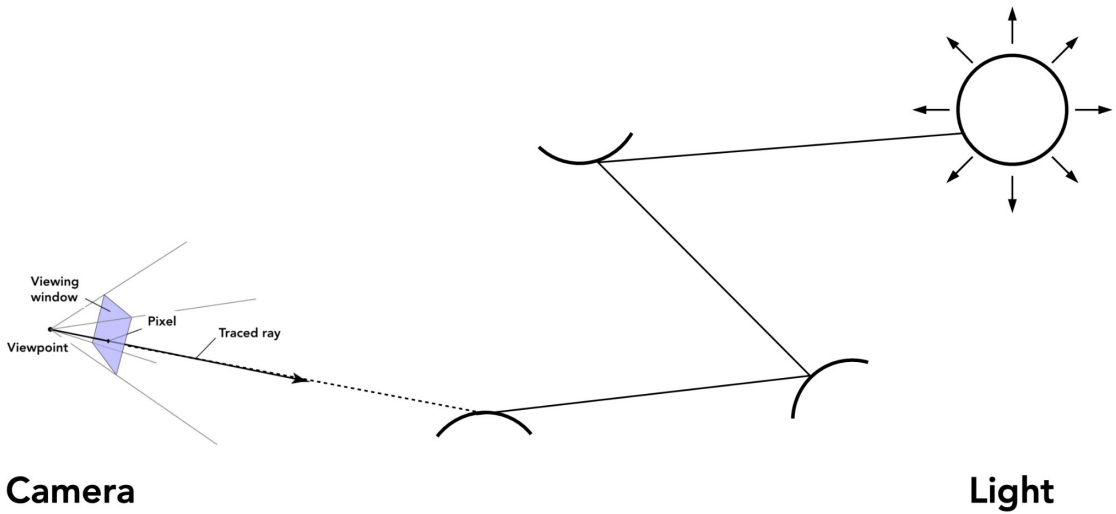
Global Illumination

- Recursively calculate how much light falls onto point *p*.



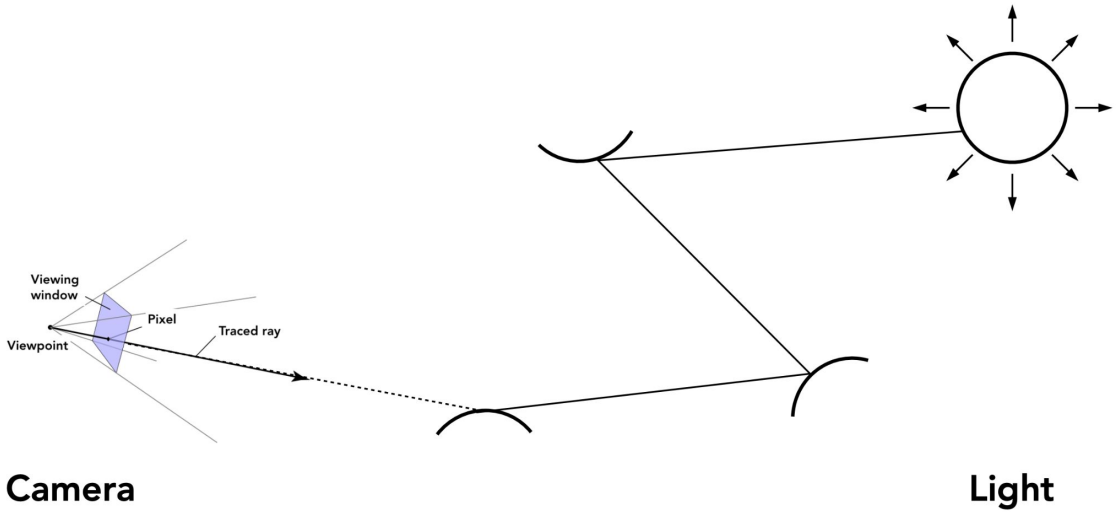
Global Illumination

- Recursively calculate how much light falls onto point *p*.
- Radiance from light source ➔ *p* ➔ camera (1-bounce).



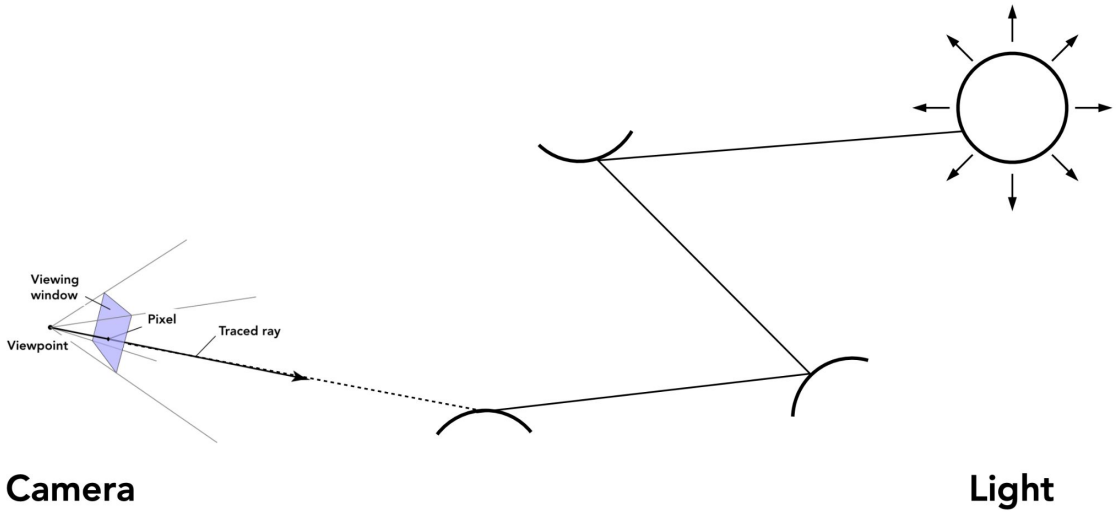
Global Illumination

- Recursively calculate how much light falls onto point *p*.
- Radiance from light source → *p* → camera (1-bounce).
- Radiance from light source → *p*’ → *p* → camera (2-bounce).



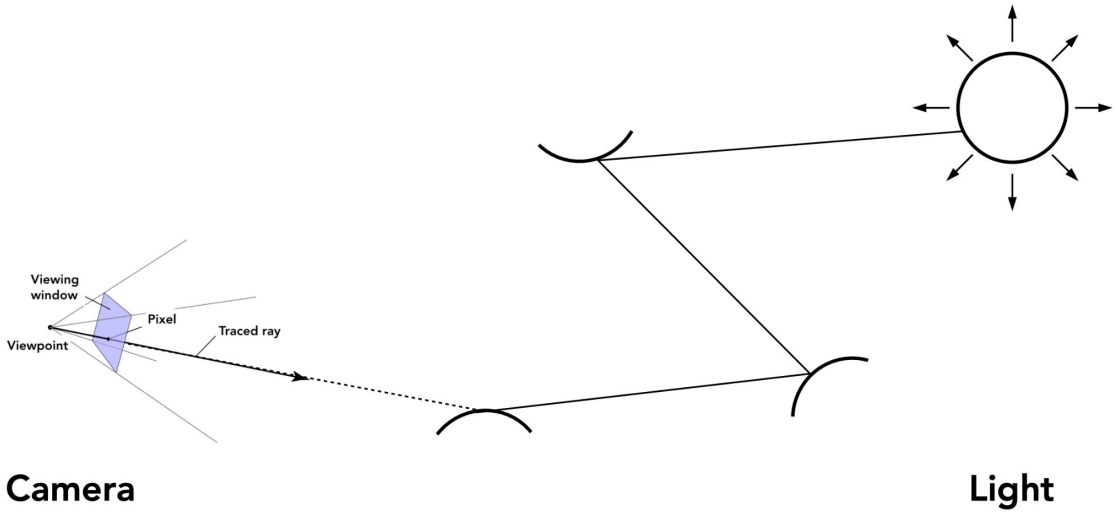
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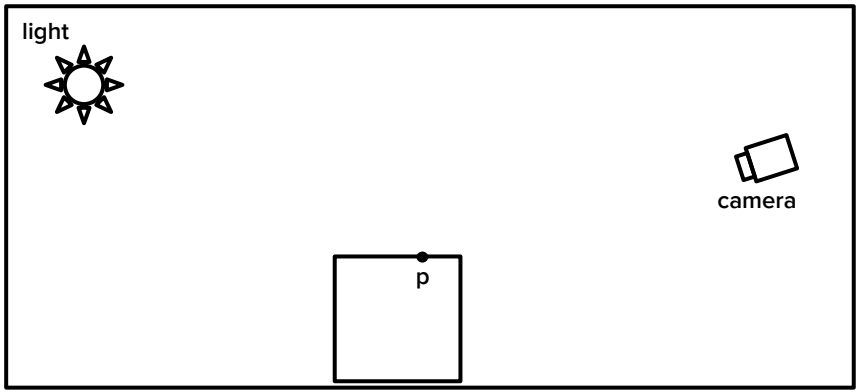
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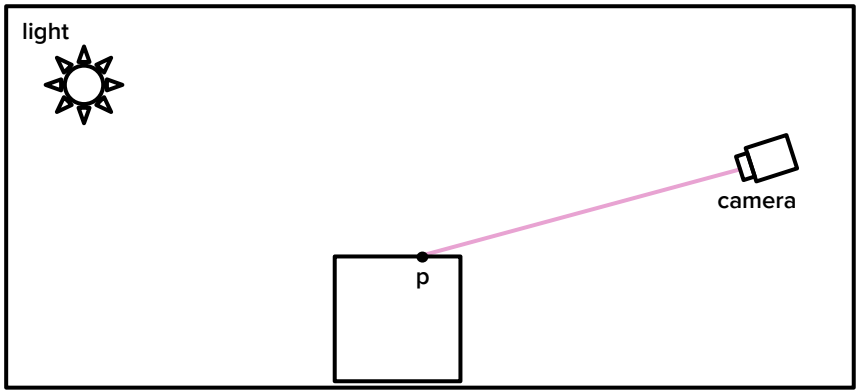


Question 4

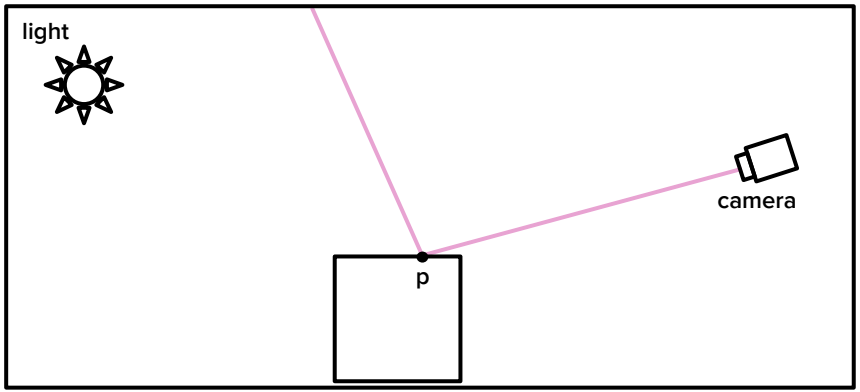
2-Bounce Path-Tracing



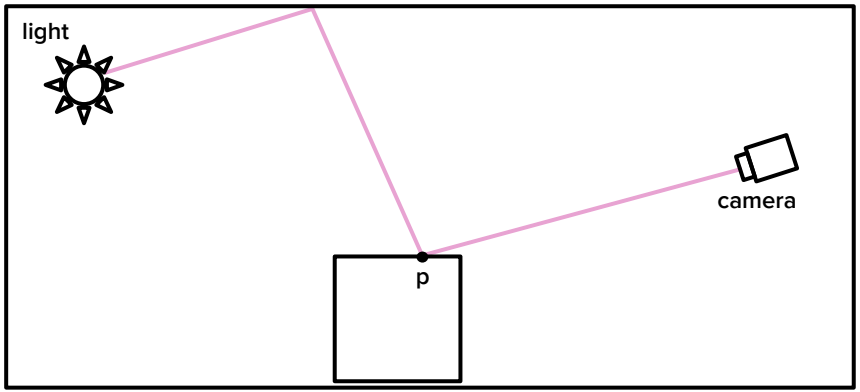
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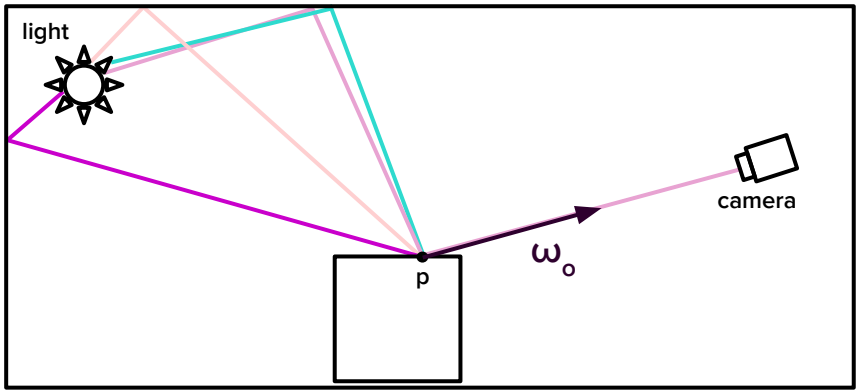
2-Bounce Path-Tracing



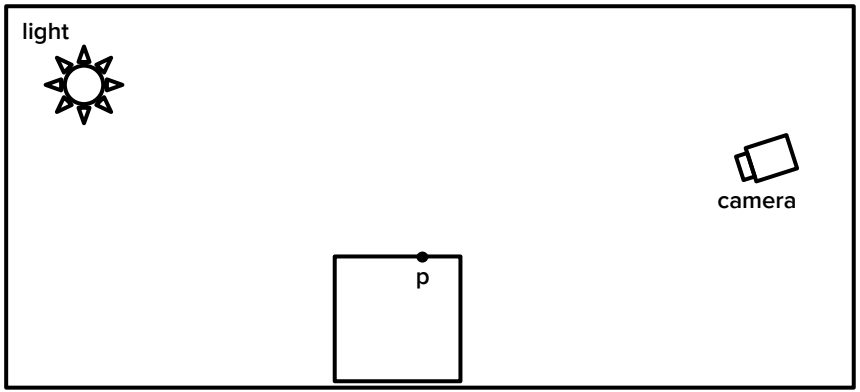
2-Bounce Path-Tracing



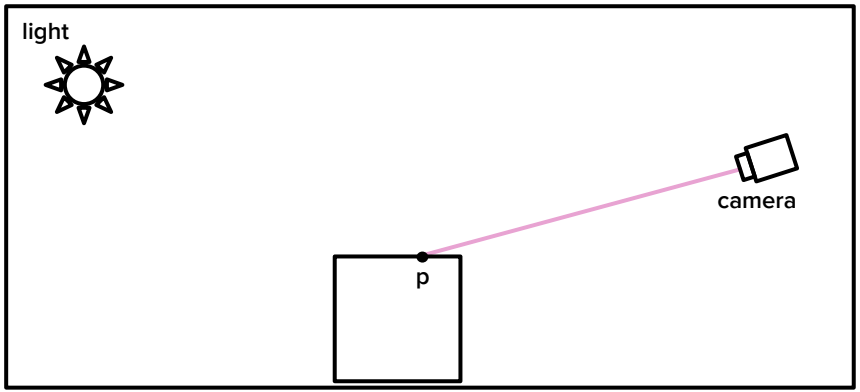
2-Bounce Path-Tracing



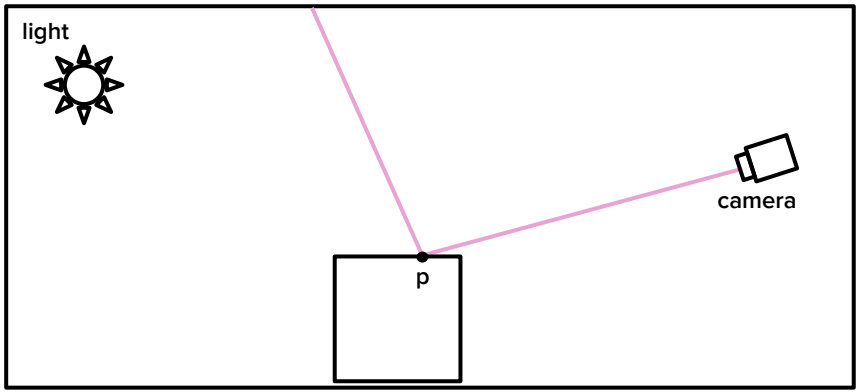
3-Bounce Path-Tracing



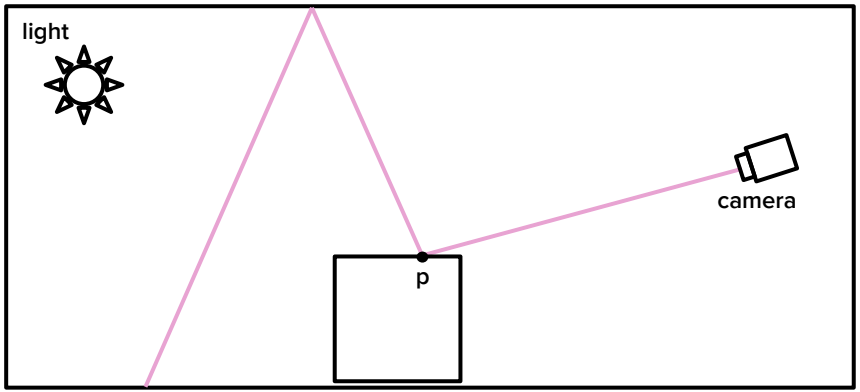
3-Bounce Path-Tracing



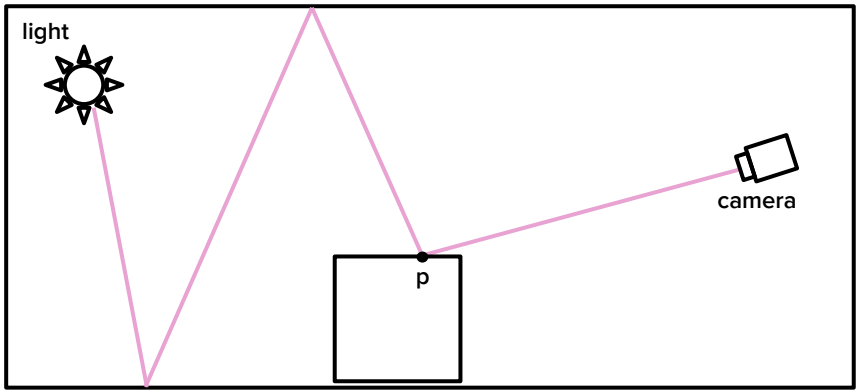
3-Bounce Path-Tracing



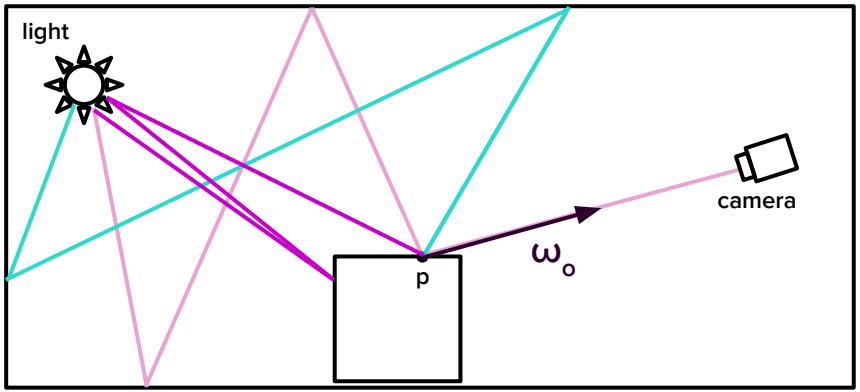
3-Bounce Path-Tracing



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Termination Conditions

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- 0-bounce, 1-bounce, 2-bounce… ➔ how to prevent ∞-bounce?

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- 0-bounce, 1-bounce, 2-bounce... ➔ how to prevent ∞-bounce?
- **Russian roulette:** at each bounce, randomly terminate current ray with probability 1 - p_{rr}.
- Unbias the estimator (proof in lecture):

$$X_{\mathrm{rr}} = \begin{cases} \dfrac{X}{p_{\mathrm{rr}}}, & \text{with probability } p_{\mathrm{rr}} \\ 0, & \text{otherwise} \end{cases}$$

Question 5